Variable Acceleration: Vectors

Questions 1–32

Question 1. A particle's displacement at time t seconds is given by $\mathbf{s}$ metres. In each case find the magnitude of the acceleration of the particle at the given value of t:

a) $\mathbf{s} = (4t^3)\mathbf{i} + (t-3)\mathbf{j}$ at $t = 0.5$

b) $\mathbf{s} = (3t^3 - 5)\mathbf{i} + (5 - t^4)\mathbf{j}$ at $t = 2$

c) $\mathbf{s} = (2e^{0.5t})\mathbf{i} + (t^2)\mathbf{j}$ at $t = 0.1$

Question 2. A particle's acceleration at time t seconds is given by $\mathbf{a}$ ms^{-2} and initial velocity $\mathbf{u}$ ms^{-1}.
In each case find the speed of the particle at the given value of t:

a) $\mathbf{a} = (4t-5)\mathbf{i} + (9t^2)\mathbf{j}, \mathbf{u} = 3\mathbf{i} - \mathbf{j}$ at $t = 5$

b) $\mathbf{a} = (12t+2)\mathbf{i} - 18\mathbf{j}, \mathbf{u} = -2\mathbf{i} + 5\mathbf{j}$ at $t = 0.5$

Question 3. A particle's velocity at time t seconds is given by $\mathbf{v}$ metres per second and initial position $\mathbf{s}$ metres. In each case find the distance of the particle from the origin at the given value of t:

a) $\mathbf{v} = (t-5)\mathbf{i} + 3\mathbf{j}, \mathbf{s} = -4\mathbf{i} + 10\mathbf{j}$ at $t = 0.4$

b) $\mathbf{v} = (3t)\mathbf{i} + (2e^{-0.1t})\mathbf{j}, \mathbf{s} = 2\mathbf{i} + 0.5\mathbf{j}$ at $t = 0.1$

Question 4. A particle P of mass 1.2 kg is moving under the action of a single force $\mathbf{F}$ newtons. At time t seconds, the velocity of P, $\mathbf{v}$ ms^{-1} is given by $\mathbf{v} = (5t+2)\mathbf{i} + (t^2 - 2t)\mathbf{j}$.
When $t = 0$, P is at the point with position vector $(2\mathbf{i} + 3\mathbf{j})$ m. Find

a) the magnitude of $\mathbf{F}$ when $t = 5$

b) the distance from its initial position when $t = 5$

Question 5. The velocity of a particle at time t seconds is $\mathbf{v} = (6t^2)\mathbf{i} + (4t - 1)\mathbf{j}$ metres per second. When $t = 1$ it passes a point A and when $t = 5$ it passes the point B. Find the distance AB.

Question 6. A particle's position vector at time t seconds is $\mathbf{r} = (2t^2 - t)\mathbf{i} + (2t + 5)\mathbf{j}$ metres.
Find the time at the instant at which the speed of the particle is 2 ms^{-1}.

Question 7. The acceleration, ms^{-2}, of a particle at time t seconds is $\mathbf{a} = (2t + 9)\mathbf{i} + (e^{-0.1t})\mathbf{j}$.
When $t = 0$ it passes a point A and when $t = 1$ it passes the point B. Show that the velocity between A and B is $10\mathbf{i} + (10 - 10e^{-0.1})\mathbf{j}$ m.

Question 8. A particle's position vector at time t seconds is $\mathbf{r} = (t^2 + t)\mathbf{i} + (5t + 4)\mathbf{j}$ metres.

a) Find the speed of the particle when $t = 4$ seconds.

b) Show that the acceleration of the particle is never zero.

Question 9. A particle's velocity at time t seconds is $\mathbf{v} = (t-4)\mathbf{i} + (t^3 + 2t)\mathbf{j}$ metres per second.
The particle is initially at the origin. Find

a) the position vector of the particle in terms of t
b) the vertical displacement when the particle has travelled 10 metres horizontally.

Question 10. A particle's velocity at time t seconds is $\mathbf{v} = (2t^{\frac{1}{2}})\mathbf{i} + (3t-1)\mathbf{j}$ metres per second.
In each case find the values of t at the instant that the particle is moving in the given direction:

a) $\mathbf{i} + \mathbf{j}$
b) $\mathbf{i} - \mathbf{j}$
c) $3\mathbf{i} + 13\mathbf{j}$

Question 11. A particle's velocity at time t s is $\mathbf{v} = (2t - 36)\mathbf{i} + (6t^{\frac{1}{2}})\mathbf{j}$ metres per second. It is known that when 1 second has passed, the position vector of the particle is $-20\mathbf{i} + 6\mathbf{j}$. Find

a) the value of t at the instant when the particle is moving in the direction $\mathbf{i} + \mathbf{j}$
b) the position vector $\mathbf{r}$ in terms of t
c) the exact position of the particle at the first instant it is moving with speed 24 ms^{-1}.

Question 12. The mass of an object is 0.5 kg. The object is moving with velocity $\mathbf{v} = (6 - pt^{\frac{1}{2}})\mathbf{i} + (5t - q)\mathbf{j}$ metres per second, where t is the number of seconds passed.
Initially, the object has position vector $4\mathbf{i} + \mathbf{j}$
After 4 seconds it passes through $12\mathbf{i} + 21\mathbf{j}$
Find

a) the values of p and q
b) the magnitude of the force acting on the object when $t = 9$ seconds.

Question 13. A particle's position $\mathbf{r}$ metres is $\mathbf{r} = (t + \frac{1}{8}k)\mathbf{i} + (kt^2 - 8t)\mathbf{j}$ where t is the time in seconds. At the instant that $t = 0.5$, the particle is moving parallel to $\mathbf{i}$. Find

a) the value of k

b) the value of t when the particle is 1 metre from the origin.

Question 14. The position vectors of particles P and Q are

$$\mathbf{r}_P = (3 - t^2)\mathbf{i} + (3t^2 + 4)\mathbf{j} \qquad \mathbf{r}_Q = (12 - 2t^2)\mathbf{i} + (4t^2 + 5)\mathbf{j}$$

respectively, where t is time in seconds.

a) Show that the distance between P and Q at any given time satisfies $d^2 = 2t^4 - 16t^2 + 82$

b) Find the time at which the distance between P and Q is minimised during the motion.

Question 15. The position vectors of particles P and Q are $\mathbf{r}_P = (t+1)\mathbf{i}+(6t)\mathbf{j}$, $\mathbf{r}_Q = (4t^2)\mathbf{i}+(5t+1)\mathbf{j}$ respectively, where t is the time in seconds.
Find the exact value of t when both particles are moving at the same speed.

Question 16. The velocity vectors of particles P and Q are $\mathbf{v}_P = (4t)\mathbf{i}+(t^2+\frac{2}{3})\mathbf{j}$, $\mathbf{v}_Q = (2t+7)\mathbf{i}+(-4)\mathbf{j}$ respectively, where t is the time in seconds.
They are both initially at the origin.
Show that particle Q is four times particle P's distance from the origin when $t = 1$

Question 17. The following questions use that i and j are unit vectors pointing east and north respectively. A particle's position **r** metres is $\mathbf{r} = (5e^{0.4t} + 1)\mathbf{i} + (t^2 - 4t)\mathbf{j}$ where t is the time in seconds.

a) Find an expression for the velocity of the particle at time t

b) State the direction in which the particle is travelling when $t = 2$

c) The mass of the particle is 10 kg. Find the magnitude of the resultant force on the particle when $t = 2$

Question 18. The mass of an object is 2 kg. The object is moving with velocity $\mathbf{v} = (t^2 + pt)\mathbf{i} + (4t^3)\mathbf{j}$ metres per second, where t is the number of seconds passed since passing through the origin. The magnitude of the force acting on the object is 120 N when $t = 2$ seconds.

a) Show that p satisfies $p^2 + 8p - 1280 = 0$

b) Hence, find the possible value of p

c) Show that there is no instantaneous time for which the object moves eastwards or westwards.

Question 19. A particle's velocity at time t seconds is $\mathbf{v} = (2t^2)\mathbf{i} + (-5t)\mathbf{j}$ metres per second.
Find the bearing the particle is travelling on when $t = 0.1$ seconds to the nearest degree.

Question 20. An object's velocity $\mathbf{v}$ metres per second is $\mathbf{v} = (-t^2 - 4)\mathbf{i} + (-t^3)\mathbf{j}$ where t is the time in seconds.
Find the time(s) at which the object is moving at a bearing of 225°

Question 21. An object's position $\mathbf{r}$ metres is given by $\mathbf{r} = (t + \frac{7}{4})\mathbf{i} + (2 - t^2)\mathbf{j}$ where t is the time in seconds. Find

a) the time(s) at which the object is at a bearing of 135° to the origin.

b) the speed of the object when it is directly east of the origin.

Question 22. A particle's position $\mathbf{r}$ metres is $\mathbf{r} = (3 + t^2)\mathbf{i} + (8t - 5)\mathbf{j}$ metres where t is the time passed in seconds. At time T, in the instant the object is moving with a bearing 045°.
Find

a) an expression for the velocity in terms of t

b) the value of T and the speed of the object at your value of T

Question 23. An object of mass 0.5 kg is initially at rest. It is being pushed by forces $\mathbf{F}_1 = -3\mathbf{i} + (t^2 - 3t + 7)\mathbf{j}$ N and $\mathbf{F}_2 = (t^2 - 1)\mathbf{i} - 5\mathbf{j}$ N where t is the number of seconds passed.
Find

a) the value of t when the object is in equilibrium.
b) the instantaneous time at which the object is moving due north-east.
c) the speed of the object when $t = 1$

Question 24. The following questions use Calculus from non-AS-Level content A particle's displacement at time t seconds is $\mathbf{s} = (2^{2t})\mathbf{i} - (2^t)\mathbf{j}$ metres. Find

a) the speed of the particle when $t = 0.4$
b) the time at which the particle is at a bearing of 135° to the origin.

Question 25. A particle's acceleration in ms^{-2} at time t seconds is $\mathbf{a} = 8\mathbf{i} - (\frac{2}{t})\mathbf{j}$.
When $t = 1$, the particle is moving with velocity $(3\mathbf{i} - 5\mathbf{j})$ ms^{-1}. Find

a) an expression for the velocity in terms of t
b) the speed of the particle at $t = 0.1$

Question 26. A particle's velocity at time t seconds is $\mathbf{v} = (e^t)\mathbf{i} + (\sin t)\mathbf{j}$ metres per second. When $t = 0$, the particle is $(2\mathbf{i} + 5\mathbf{j})$ m from the origin.
Find the distance of the particle from the origin when $t = 0.5$

Question 27. A particle's position in relation to the origin is $\mathbf{r} = (4\cos(2t))\mathbf{i} + (t^2 \sin t)\mathbf{j}$ metres where t is the time in seconds. The mass of the particle is 0.5 kg.

a) State the position of the particle in relation to the origin when $t = 3\pi$

b) Find the direction in which the particle is travelling when $t = 3\pi$

c) Find the magnitude of the resultant force on the particle when $t = 3\pi$

Question 28. A particle's acceleration at time t is $\mathbf{a} = 6\mathbf{i} + (te^{2t})\mathbf{j}$ metres per second squared.
The initial velocity of the particle is $(\frac{3}{4}\mathbf{j})$ ms^{-1}.
Find the speed of the particle when $t = 0.2$ seconds.

Question 29. The velocity of a particle at time t seconds is $\mathbf{v} = (\sin 2t)\mathbf{i} + (\cos t)\mathbf{j}$ metres per second. When $t = 0$ it passes a point A and when $t = \frac{\pi}{2}$ it passes the point B.
Find the distance AB.

Question 30. A particle's position at time t seconds is $\mathbf{r} = (\cos t - e^t)\mathbf{i} + (\sin t - \ln t)\mathbf{j}$ metres.
Find the bearing the particle is travelling on when $t = 0.5$ seconds to the nearest degree.

Question 31. Prerequisite knowledge for Q31–Q32: Parametric Equations For each of the following, write a cartesian equation for the path of the motion:

a) $\mathbf{r} = (8t - 5)\mathbf{i} + (3 + t^2)\mathbf{j}$
b) $\mathbf{r} = (t + \frac{7}{4})\mathbf{i} + (2 - t^2)\mathbf{j}$
c) $\mathbf{r} = (t^2)\mathbf{i} + (5t + 4)\mathbf{j}$

Question 32. A particle P has mass 2 kg and its position is given by $\mathbf{r}$ metres $\mathbf{r} = (\sin t)\mathbf{i} + (\cos t)\mathbf{j}$ where t is the time in seconds.

a) Find the magnitude of the resultant force on the particle when $t = \frac{\pi}{2}$
b) Write a cartesian equation for the path of the motion.